

DN14292 Chueng-Ryong Ji and Hariprashad Ravikumar Interpolating conformal algebra in $(1 + 1)$ dimensions between the instant form and the light-front form of relativistic dynamics

The authors present a technically strong construction of an interpolating conformal algebra in $(1+1)$ dimensions connecting the instant and light-front forms of relativistic dynamics. The analysis clarifies the structure of the generators and their representations, in particular the redistribution between dynamical and kinematical generators, leading to a maximal set of kinematical generators in the light-front formulation.

The formal development is careful and mathematically consistent. The authors indicate a possible extension to the $(3+1)$ -dimensional conformal group with applications to QCD and hadron physics. To strengthen the physical motivation, however, the manuscript would benefit from a more concrete discussion of the specific physical problems this framework is expected to address.

I recommend publication in PRD, subject to clarification on this point.